

Probability Theory

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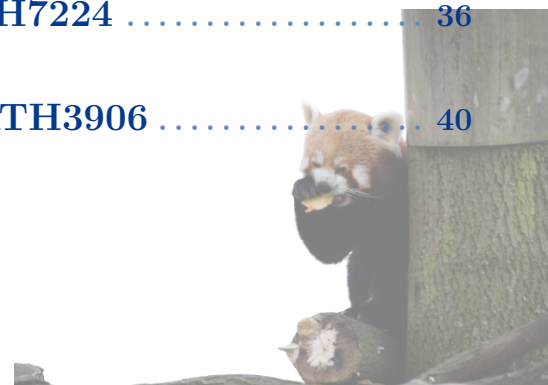
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Part I: MATH3603

1 Discrete Probability Distribution

1.1 Introduction

Definition. The **sample space** is the set of all possible outcomes of a random experiment. If Ω is the sample space, then

$$\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$$

An **event** is any subset of the sample space. If $A \subseteq \Omega$, we say that A is an event.

Notation. Let X be a random variable and let $a \in \mathbb{R}$. The notation

$$\{X = a\}$$

denotes the event

$$\{\omega \in \Omega : X(\omega) = a\}$$

Definition. The probability of an event A is a number between 0 and 1 that represents the likelihood of event A occurring. It is denoted by $\mathbb{P}(A)$ and satisfies the following axioms:

- $\mathbb{P}(\Omega) = 1$, meaning the probability of the entire sample space is 1.
- $\mathbb{P}(A) \geq 0$ for all events A .
- If $A_1, A_2, \dots, A_n$ are disjoint events, then $\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n \mathbb{P}(A_i)$ where

$$\bigcup_{i=1}^n A_i = \{x \mid x \in A_i \text{ for some } i \in \{1, 2, \dots, n\}\}$$

Theorem. (Theoretical Probability)

$$\mathbb{P}(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$



Theorem. The complement of event A , denoted by A^c , represents the event that A does not occur. The probability of the complement is given by

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$$

Theorem. The union of two events A and B , denoted by $A \cup B$, represents the event that either A , B , or both occur. The probability of the union is

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

Theorem. The conditional probability of event A given event B , denoted by $\mathbb{P}(A \mid B)$, is the probability that A occurs given that B has already occurred. It is given by

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \quad \text{if } \mathbb{P}(B) > 0$$

More generally, for events $A_1, A_2, \dots, A_n$,

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i\right) = \mathbb{P}(A_1) \prod_{i=2}^n \mathbb{P}(A_i \mid A_1 \cap \dots \cap A_{i-1})$$

Proof. When we are given that B has occurred, the new sample space becomes B . Within this restricted sample space, the probability of A should be the proportion of outcomes in B that also belong to A . Hence,

$$\mathbb{P}(A \mid B) = \frac{\text{Probability of outcomes in } A \cap B}{\text{Probability of outcomes in } B} = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

Theorem. If A and B are independent events, then

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$$

Theorem. (Bayes' Theorem) Bayes' theorem relates conditional probabilities and is given by

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A) \cdot \mathbb{P}(A)}{\mathbb{P}(B)}$$



Definition. A **random variable** is a function

$$X : \Omega \rightarrow \mathbb{R}$$

where Ω is the sample space of a random experiment. A random variable X is called **discrete** if its set of possible values

$$\{x_1, x_2, x_3, \dots\}$$

is finite or countably infinite.

Definition. Let X be a discrete random variable. The **probability mass function (PMF)** of X is defined by

$$p_X(x) = \mathbb{P}(X = x)$$

For a valid PMF, the following must hold:

- $p_X(x) \geq 0$ for all x
- $\sum_x p_X(x) = 1$

Definition. The **cumulative distribution function (CDF)** of a random variable X is

$$F_X(x) = \mathbb{P}(X \leq x)$$

For a discrete random variable,

$$F_X(x) = \sum_{t \leq x} p_X(t)$$

Definition. The **expected value** of a random variable X , denoted $\mathbb{E}[X]$, is the long-run average value of X . For discrete random variables, it is calculated as

$$\mathbb{E}[X] = \sum_x x p_X(x)$$

Theorem. (Linearity of Expectation) For constants $a, b \in \mathbb{R}$,

$$\mathbb{E}[aX + bY] = a \mathbb{E}[X] + b \mathbb{E}[Y]$$



By repeatedly applying this theorem. we can note that

$$\mathbb{E}\left[\sum_{i=1}^n a_i X_i\right] = \sum_{i=1}^n a_i \mathbb{E}[X_i]$$

where $X_1, X_2, \dots, X_n$ are random variables with finite expectations, and $a_1, a_2, \dots, a_n \in \mathbb{R}$.

Definition. The **variance** of a random variable X , denoted $\text{Var}(X)$, measures the spread of the values of X around the expected value. It is calculated as

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Properties. If $c \in \mathbb{R}$ is a constant, then

$$\text{Var}(c) = 0$$

Properties. For constants $a, b \in \mathbb{R}$,

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Properties. For any random variables X and Y ,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$$

where the covariance is defined by

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

If X and Y are independent, then

$$\text{Cov}(X, Y) = 0$$

and hence

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

Theorem. Let X and Y be discrete random variables. The **conditional expectation of X given $Y = y$** is defined as

$$\mathbb{E}(X \mid Y = y) = \sum_x x \mathbb{P}(X = x \mid Y = y)$$



provided $\mathbb{P}(Y = y) > 0$.

Theorem. (Towering Property) Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let X and Y be random variables such that

$$E[|X|] < \infty$$

Then

$$E(E(X | Y)) = E[X]$$



1.2 Bernoulli Distribution

Definition. Many random experiments have only two possible outcomes: **success** or **failure**. For example, tossing a biased coin (heads or tails). A random variable X is said to follow a **Bernoulli distribution** with parameter $p \in (0, 1)$, written

$$X \sim \text{Bernoulli}(p)$$

if

$$\mathbb{P}(X = 1) = p, \quad \mathbb{P}(X = 0) = 1 - p$$

where $X = 1$ represents success and $X = 0$ represents failure.

Theorem. If $X \sim \text{Bernoulli}(p)$, then

$$\mathbb{E}[X] = p, \quad \text{Var}(X) = p(1 - p)$$

Proof. By definition,

$$\mathbb{E}[X] = 0 \cdot (1 - p) + 1 \cdot p = p$$

Since $X^2 = X$ for $X \in \{0, 1\}$,

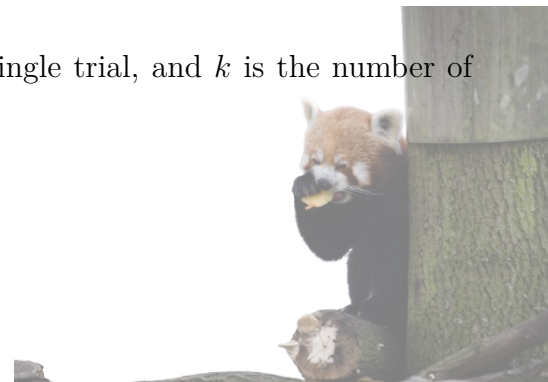
$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1 - p)$$

1.3 Binomial Distribution

Definition. (Binomial Distribution) The binomial distribution is a discrete probability distribution that describes the number of successes in a fixed number of independent Bernoulli trials, each with the same probability of success. A random variable X is said to follow a binomial distribution with parameters n and p , denoted as $X \sim \text{Binomial}(n, p)$, if its probability mass function is given by

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, 2, \dots, n$$

where n is the number of trials, p is the probability of success on a single trial, and k is the number of successes in n trials.



Proof. Consider a sequence of n trials. Each trial can result in either success or failure. We need to calculate the number of ways in which exactly k successes occur. The number of ways to choose k successes out of n trials is given by the binomial coefficient $\binom{n}{k}$. This represents the number of distinct sequences that contain exactly k successes. For any given sequence of k successes and $n - k$ failures, the probability of that sequence occurring is the product of the probabilities of individual trials:

$$p^k(1 - p)^{n-k}$$

because the probability of success is p and the probability of failure is $1 - p$. As the trials are independent, the total probability of exactly k successes in n trials is the product of the number of ways to choose k successes and the probability of each specific sequence:

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Theorem. The mean or expected value of X is

$$\mathbb{E}(X) = np$$

The variance of X is

$$\text{Var}(X) = np(1 - p)$$



Proof. Let

$$X = X_1 + X_2 + \cdots + X_n$$

where

$$X_i = \begin{cases} 1, & \text{if the } i\text{-th trial is a success} \\ 0, & \text{otherwise} \end{cases}$$

Each X_i is a Bernoulli random variable with

$$P(X_i = 1) = p, \quad P(X_i = 0) = 1 - p$$

The expectation of each X_i is

$$\mathbb{E}[X_i] = 1 \cdot p + 0 \cdot (1 - p) = p$$

By the linearity of expectation,

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbb{E}[X_i] = \sum_{i=1}^n p = np$$

For the variance of each X_i ,

$$\text{Var}(X_i) = \mathbb{E}(X_i^2) - [\mathbb{E}(X_i)]^2$$

Since $X_i^2 = X_i$,

$$\mathbb{E}[X_i^2] = \mathbb{E}[X_i] = p$$

Hence,

$$\text{Var}(X_i) = p - p^2 = p(1 - p)$$

As the trials are independent, the random variables $X_1, \dots, X_n$ are independent. Therefore,

$$\text{Var}(X) = \sum_{i=1}^n \text{Var}(X_i)$$

Thus,

$$\text{Var}(X) = \sum_{i=1}^n p(1 - p) = np(1 - p)$$



1.4 Poisson Distribution

Definition. The **Poisson distribution** is a discrete probability distribution that models the number of times an event occurs in a fixed interval of time or space, under the following assumptions:

- Events occur independently.
- The average rate of occurrence is constant.
- Two events cannot occur at exactly the same instant.

Typical applications include:

- Number of radioactive decays in a given time.
- Number of emails received per hour.
- Number of defects per unit length of material.

Theorem. A discrete random variable X is said to follow a **Poisson distribution** with parameter $\lambda > 0$, written

$$X \sim \text{Poisson}(\lambda)$$

if its probability mass function is given by

$$\mathbb{P}(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

Here, λ represents the expected number of occurrences in the given interval.

Proposition. The Poisson distribution arises as a limiting case of the Binomial distribution when events are rare but trials are frequent.



Proof. Let $X_n \sim \text{Binomial}(n, p_n)$, where

$$p_n = \frac{\lambda}{n}$$

We will show that as $n \rightarrow \infty$, the distribution of X_n converges to a $\text{Poisson}(\lambda)$ distribution. The binomial probability mass function is

$$\mathbb{P}(X_n = k) = \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

Note that

$$\binom{n}{k} = \frac{n(n-1) \cdots (n-k+1)}{k!}$$

Therefore,

$$\mathbb{P}(X_n = k) = \frac{n(n-1) \cdots (n-k+1)}{n^k} \cdot \frac{\lambda^k}{k!} \cdot \left(1 - \frac{\lambda}{n}\right)^n \cdot \left(1 - \frac{\lambda}{n}\right)^{-k}$$

Take limits term by term as $n \rightarrow \infty$:

- $\frac{n(n-1) \cdots (n-k+1)}{n^k} \rightarrow 1$
- $\left(1 - \frac{\lambda}{n}\right)^n \rightarrow e^{-\lambda}$
- $\left(1 - \frac{\lambda}{n}\right)^{-k} \rightarrow 1$

Hence,

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$



1.5 Moment Generating Function

Definition. Let Ω be a non-empty set (called the **sample space**). A collection $\mathcal{F} \subseteq 2^\Omega$ is called a **σ -algebra** on Ω if

- $\Omega \in \mathcal{F}$,
- If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$, and
- If $A_1, A_2, A_3, \dots \in \mathcal{F}$, then

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$$

where

$$2^\Omega = \{A \mid A \subseteq \Omega\}$$

On $\mathbb{R}$, the **Borel σ -algebra** $\mathcal{B}(\mathbb{R})$ is the smallest sigma-algebra containing all open intervals.

Definition. Let X be a real-valued random variable. The **moment generating function** of X is defined by

$$M_X(t) = \mathbb{E} [e^{tX}]$$

for all t in an open interval containing 0 where the expectation exists.

Example. Consider the probability space $(S, \mathcal{B}, P)$ where $S = (0, 6)$, $\mathcal{B}$ is the Borel σ -algebra, and P is the uniform distribution. A random variable $X : S \rightarrow \mathbb{R}$ is defined by

$$X(s) = \begin{cases} 1, & 0 < s < 2 \\ 2, & 2 \leq s < 5 \\ -1, & 5 \leq s < 6 \end{cases}$$

- What is the range of X ?
- What is the event $\{X < 0\}$ and what is its probability?
- What is the moment generating function of X ?
- Are X and $\mathbf{1}_{(2,4)}$ independent random variables? Here

$$\mathbf{1}_A(x) = \begin{cases} 1, & \text{if } x \in A \\ 0, & \text{if } x \notin A \end{cases}$$



Solution.

(a) From the definition of X , the only values taken by X are

$$-1, 1, 2$$

Hence, the range of X is $\{-1, 1, 2\}$.

(b) The event $\{X < 0\}$ corresponds to

$$\{s \in S : X(s) = -1\} = [5, 6)$$

Since P is the uniform distribution on $(0, 6)$,

$$P(X < 0) = \frac{\text{length of } [5, 6)}{\text{length of } (0, 6)} = \frac{1}{6}$$

(c) The moment generating function of X is

$$M_X(t) = \mathbb{E}[e^{tX}] = \sum_x e^{tx} P(X = x)$$

where

$$P(X = 1) = \frac{2}{6} = \frac{1}{3}, \quad P(X = 2) = \frac{3}{6} = \frac{1}{2}, \quad P(X = -1) = \frac{1}{6}$$

Therefore,

$$M_X(t) = \frac{1}{3}e^t + \frac{1}{2}e^{2t} + \frac{1}{6}e^{-t}$$

(d) Let $Y = \mathbf{1}_{(2,4)}$. Consider the events

$$\{X = 1\} = (0, 2), \quad \{Y = 1\} = (2, 4)$$

Then

$$P(X = 1) = \frac{2}{6} = \frac{1}{3}, \quad P(Y = 1) = \frac{2}{6} = \frac{1}{3}$$

but

$$P(X = 1, Y = 1) = P((0, 2) \cap (2, 4)) = 0$$



Since

$$P(X = 1, Y = 1) \neq P(X = 1)P(Y = 1)$$

X and $\mathbf{1}_{(2,4)}$ are **not independent**.

1.6 Pólya's Urn Model

Definition. A random variable X is said to have a **Beta distribution** with parameters $\alpha > 0$ and $\beta > 0$, written

$$X \sim \text{Beta}(\alpha, \beta)$$

if its probability density function is

$$f_X(x) = \begin{cases} \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

where $B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx$ is the **Beta function**.

Definition. An urn initially contains:

- $a > 0$ red balls,
- $b > 0$ blue balls.

At each discrete time step:

- A ball is drawn uniformly at random from the urn.
- The ball is returned to the urn together with **one additional ball of the same color**.

This process is repeated indefinitely.

Proposition. Let X_n be the indicator variable that the n th draw is red. Let

$$\overline{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

As $n \rightarrow \infty$,

$$\overline{X}_n \xrightarrow{\text{a.s.}} Y$$

where Y has a **Beta distribution**:

$$Y \sim \text{Beta}\left(\frac{r}{c}, \frac{b}{c}\right)$$



Definition. In Bose–Einstein statistics, particles are

- indistinguishable, and
- allowed to occupy the same state without restriction.



2 Continuous Probability Distribution

2.1 Introduction

Definition. A **continuous random variable** is a random variable that can take **any value in an interval of real numbers**.

Proposition. A random variable X is said to be **continuous** if there exists a function $f_X(x)$ such that for any $a < b$,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx$$

The function $f_X(x)$ is called the **probability density function (probability density function)** of X .

Proposition. A function $f_X(x)$ is a valid probability density function if and only if

- $f_X(x) \geq 0$ for all $x \in \mathbb{R}$

-

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

Proposition. For a continuous random variable,

$$\mathbb{P}(X = a) = 0 \quad \text{for all } a \in \mathbb{R}$$

Proof.

$$\mathbb{P}(X = a) = \int_a^a f_X(x) dx = 0$$

Proposition. If $F_X(x)$ is differentiable, then

$$f_X(x) = \frac{d}{dx} F_X(x)$$

Definition. The **expected value** (mean) of a continuous random variable X is

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$



provided the integral converges.

Theorem. Note that for any function $g(X)$,

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x) dx$$

Definition. A random variable X has a **uniform distribution** on $[a, b]$ if

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise} \end{cases}$$

Theorem. For uniform distribution,

$$\mathbb{E}[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}$$

Definition. Let X and Y be two continuous random variables. A function $f(x, y)$ is called the **joint probability density function** of (X, Y) if for any region $R \subset \mathbb{R}^2$,

$$\mathbb{P}((X, Y) \in R) = \iint_R f(x, y) dx dy.$$

Definition. The **marginal probability density function** of X is obtained by integrating out Y :

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

Similarly, the marginal probability density function of Y is

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

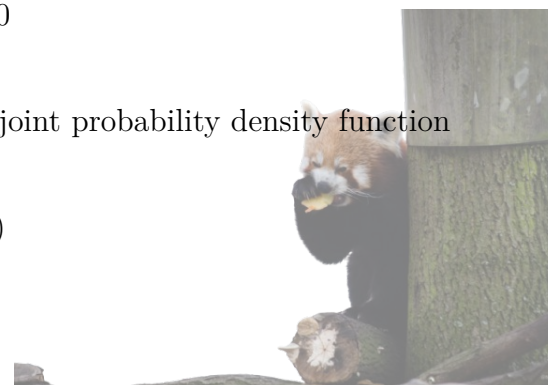
Definition Let X and Y be continuous random variables with joint probability density function $f_{X,Y}(x, y)$.

The **conditional density of X given $Y = y$** is defined as

$$f_{X|Y}(x | y) = \frac{f(x, y)}{f_Y(y)}, \quad \text{for } f_Y(y) > 0$$

Example. Let X and Y be jointly continuous random variables with joint probability density function

$$f(x, y) = \frac{2}{5}(x^3y + 2x) \mathbf{1}_{(0,1) \times (0,2)}(x, y)$$



Find $E(Y | X)$ and $E(Y(X^2 + 2))$ in terms of $\mu = \mathbb{E}(X)$ and $\sigma^2 = \text{Var}(X)$

Solution. First compute the marginal density of X :

$$f_X(x) = \int_0^2 \frac{2}{5}(x^3 y + 2x) dy = \frac{2}{5} \left(x^3 \int_0^2 y dy + 2x \int_0^2 dy \right)$$

Since $\int_0^2 y dy = 2$ and $\int_0^2 dy = 2$, we obtain

$$f_X(x) = \frac{4}{5}(x^3 + 2x), \quad 0 < x < 1$$

The conditional density of Y given $X = x$ is

$$f_{Y|X}(y | x) = \frac{f(x, y)}{f_X(x)} = \frac{x^3 y + 2x}{2(x^3 + 2x)}, \quad 0 < y < 2$$

Hence,

$$E(Y | X = x) = \int_0^2 y f_{Y|X}(y | x) dy = \frac{1}{2(x^3 + 2x)} \int_0^2 (x^3 y^2 + 2xy) dy.$$

Using $\int_0^2 y^2 dy = \frac{8}{3}$ and $\int_0^2 y dy = 2$, we obtain

$$E(Y | X = x) = \frac{\frac{8}{3}x^3 + 4x}{2(x^3 + 2x)} = \frac{4x^2 + 6}{3(x^2 + 2)}$$

By the tower property,

$$E(Y(X^2 + 2)) = E[(X^2 + 2)E(Y | X)]$$

Substituting the result from part (a),

$$(X^2 + 2)E(Y | X) = \frac{4X^2 + 6}{3}$$

Therefore,

$$E(Y(X^2 + 2)) = \frac{1}{3}E(4X^2 + 6) = \frac{4}{3}E(X^2) + 2.$$

As $E(X^2) = \sigma^2 + \mu^2$, we obtain

$$E(Y(X^2 + 2)) = \frac{4}{3}(\sigma^2 + \mu^2) + 2$$



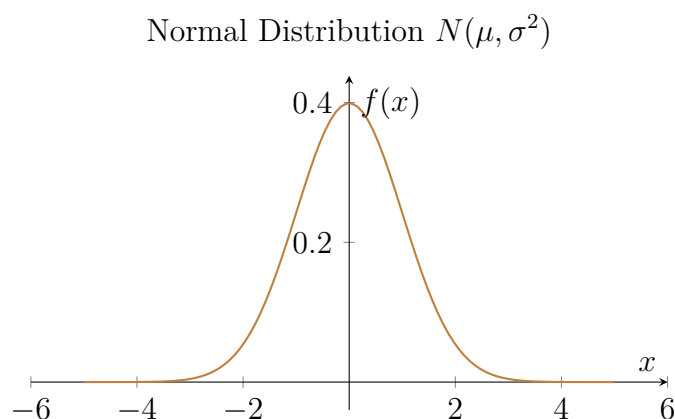
2.2 Normal Distribution

Definition. The **normal distribution** is a continuous probability distribution defined by the probability density function

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where μ is the mean and σ is the standard deviation.

Properties. The normal distribution is symmetric and bell-shaped. The following plot represents a normal distribution with $\mu = 0$ and $\sigma = 1$ (standard normal distribution). The standard normal variable is often denoted by Z .



Properties.

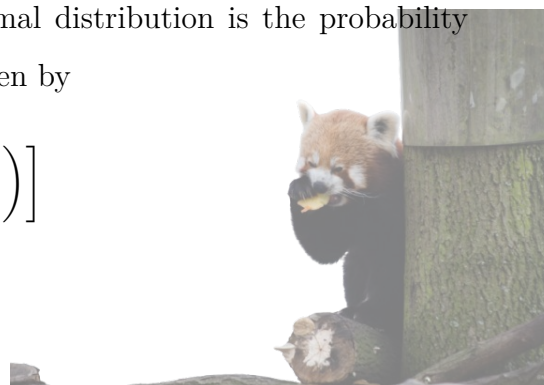
- The mean μ , median, and mode of a normal distribution are all equal.
- The total area under the probability density function is 1, which means

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

- It follows the empirical rule (68-95-99.7 rule): About 68% of the data falls within one standard deviation ($\mu \pm \sigma$). About 95% of the data falls within two standard deviations ($\mu \pm 2\sigma$). About 99.7% of the data falls within three standard deviations ($\mu \pm 3\sigma$).

Proposition. The cumulative distribution function (CDF) of a normal distribution is the probability that the random variable X is less than or equal to a value x . It is given by

$$F_X(x) = \mathbb{P}(X \leq x) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{x - \mu}{\sigma\sqrt{2}} \right) \right]$$



where $\text{erf}(z)$ is the error function, defined as

$$\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z \exp(-t^2) dt$$

Proposition. A **standard normal distribution table**, or **Z-table**, is used to find probabilities of the form

$$\mathbb{P}(Z \leq z), \quad \mathbb{P}(Z \geq z), \quad \mathbb{P}(a \leq Z \leq b)$$

Most standard normal tables give the value

$$\Phi(z) = \mathbb{P}(Z \leq z)$$

which is the **area under the standard normal curve to the left of z** . In the table,

- The **row** gives the first decimal place of z (e.g. 1.2).
- The **column** gives the second decimal place of z (e.g. 0.03).
- The table entry is $\Phi(z)$.

Proposition. The standard normal distribution is symmetric about 0:

$$\mathbb{P}(Z \leq -z) = 1 - \mathbb{P}(Z \leq z)$$

2.3 Strong Law of Large Numbers

Let $\{X_n\}_{n=1}^\infty$ be independent and identically distributed random variables such that

$$\mathbb{E}[X_1] = \mu, \quad \mathbb{E}[|X_1|] < \infty$$

Then

$$\bar{X}_n \xrightarrow{\text{a.s.}} \mu$$

which is

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1$$

Interpretation: The observed average of a large number of experiments eventually stabilizes at the expected value.



2.4 Central Limit Theorem

Theorem. Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables with mean μ and variance σ^2 . The sample mean is given by

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

The Central Limit Theorem states that as $n \rightarrow \infty$, the distribution of $\bar{X}_n$ approaches a normal distribution:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1)$$

2.5 Poisson Process

Definition. A continuous random variable X is said to follow an **exponential distribution** with rate parameter $\lambda > 0$, denoted by

$$X \sim \text{Exp}(\lambda)$$

if its probability density function is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

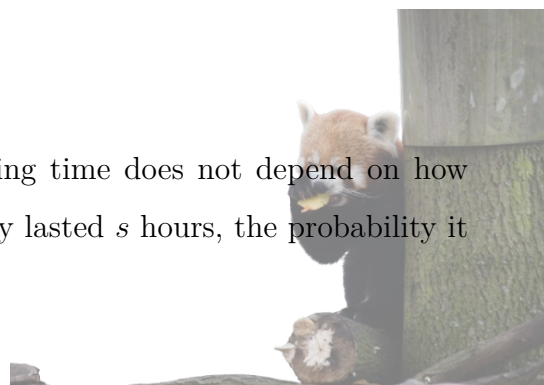
The cumulative distribution function is

$$F_X(x) = \mathbb{P}(X \leq x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

The exponential distribution is commonly used to model **waiting times** until an event occurs, such as:

- Time until a radioactive particle decays,
- Time until the next customer arrives,
- Lifetime of a component with constant failure rate.

Proposition. The memoryless property means that the future waiting time does not depend on how much time has already elapsed. For example, if a light bulb has already lasted s hours, the probability it



lasts at least t more hours is the same as if it were brand new.

Theorem. Let $X \sim \text{Exp}(\lambda)$. Then

$$\mathbb{E}[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}$$

Example. Let X be a Bernoulli random variable of which the probability of success is e^{-Y} , i.e.

$$X \sim \text{Bernoulli}(e^{-Y})$$

where

$$Y \sim \text{Exp}(\lambda)$$

- (a) Find the probability mass function of X .
- (b) Find the conditional cumulative distribution function of Y given that $X = 1$.

Solution. Given $Y = y$, the conditional distribution of X is

$$P(X = 1 \mid Y = y) = e^{-y}, \quad P(X = 0 \mid Y = y) = 1 - e^{-y}$$

The marginal probability that $X = 1$ is obtained by taking expectation with respect to Y :

$$P(X = 1) = \mathbb{E}[e^{-Y}]$$

As $Y \sim \text{Exp}(\lambda)$, its density is

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y \geq 0$$

Then

$$\begin{aligned} P(X = 1) &= \int_0^\infty e^{-y} \lambda e^{-\lambda y} dy \\ &= \lambda \int_0^\infty e^{-(\lambda+1)y} dy \\ &= \frac{\lambda}{\lambda+1} \end{aligned}$$

Therefore,

$$P(X = 0) = 1 - P(X = 1) = \frac{1}{\lambda+1}$$



Hence, the probability mass function of X is

$$P(X = x) = \begin{cases} \frac{\lambda}{\lambda + 1}, & x = 1 \\ \frac{1}{\lambda + 1}, & x = 0 \end{cases}$$

Example. In order to boost sales in a shopping mall, customers can take their purchase receipts (over certain amount of expenditure) to exchange for some shopping coupons in the mall. Peter, Queenie and Robert arrive at the exchange point simultaneously. Since there are two counters only, Peter and Queenie go directly into service while Robert waits for the first available counter. Each of them will leave the exchange point immediately after collecting the coupons. Suppose that the two counters have exponential service times with parameters $1/3$ and $1/6$ customers per minute respectively.

Let $X \sim \text{Exp}(1/3)$ and $Y \sim \text{Exp}(1/6)$ be the service times at the two counters. They are independent. Robert enters service when the first of the two services is completed. Hence his waiting time is

$$W = \min(X, Y)$$

Claim:

$$W \sim \text{Exp}\left(\frac{1}{2}\right)$$

Proof. For any $t \geq 0$,

$$P(W > t) = P(\min(X, Y) > t)$$

The event $\{\min(X, Y) > t\}$ occurs if and only if both $X > t$ and $Y > t$. Hence,

$$P(W > t) = P(X > t, Y > t)$$

As X and Y are independent,

$$P(X > t, Y > t) = P(X > t) P(Y > t)$$

For an exponential random variable $Z \sim \text{Exp}(\lambda)$,

$$P(Z > t) = e^{-\lambda t}, \quad t \geq 0$$



Therefore,

$$\begin{aligned} P(W > t) &= e^{-(1/3)t} \cdot e^{-(1/6)t} \\ &= e^{-(1/3+1/6)t} \\ &= e^{-(1/2)t} \end{aligned}$$

The survival function of an exponential random variable with rate λ is

$$P(Z > t) = e^{-\lambda t}$$

Since

$$P(W > t) = e^{-(1/2)t}$$

we conclude that

$$W \sim \text{Exp}\left(\frac{1}{2}\right)$$

Therefore,

$$\mathbb{E}[W] = \frac{1}{1/2} = 2 \text{ minutes}$$

Using Bayes' theorem, the conditional density of Y given $X = 1$ is

$$f_{Y|X=1}(y) = \frac{P(X = 1 | Y = y)f_Y(y)}{P(X = 1)}$$

Substituting the known expressions,

$$\begin{aligned} f_{Y|X=1}(y) &= \frac{e^{-y} \cdot \lambda e^{-\lambda y}}{\lambda/(\lambda + 1)} \\ &= (\lambda + 1)e^{-(\lambda+1)y}, \quad y \geq 0 \end{aligned}$$

Hence, $Y | X = 1 \sim \text{Exp}(\lambda + 1)$. The conditional cumulative distribution function is therefore

$$F_{Y|X=1}(y) = P(Y \leq y | X = 1) = \begin{cases} 1 - e^{-(\lambda+1)y}, & y \geq 0 \\ 0, & y < 0 \end{cases}$$



Definition. A non-negative random variable X is said to be **memoryless** if for all $s, t \geq 0$,

$$\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)$$

Example. Exponential distribution is memoryless.

Solution. Let $X \sim \text{Exp}(\lambda)$. Then

$$\mathbb{P}(X > x) = e^{-\lambda x}, \quad x \geq 0$$

Hence,

$$\mathbb{P}(X > s + t \mid X > s) = \frac{\mathbb{P}(X > s + t)}{\mathbb{P}(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = \mathbb{P}(X > t)$$

Theorem. The exponential distribution is the **only continuous distribution** on $[0, \infty)$ that satisfies the memoryless property.

Definition. A **stochastic process** is a family of random variables

$$\{X(t) : t \in T\}$$

defined on the same probability space, where

- t is the **index** (usually time), and
- T may be discrete or continuous

The following gives the example of stochastic process:

Time Index	State Space	Example
Discrete	Discrete	Markov chain
Discrete	Continuous	Random walk
Continuous	Discrete	Poisson process
Continuous	Continuous	Brownian motion

Definition. $\{N(t)\}_{t \geq 0}$ is a **counting process** where $N(t)$ counts the number of events occurring in $[0, t]$.

Definition. A stochastic process $\{N(t)\}_{t \geq 0}$ is called a **Poisson process with rate $\lambda > 0$** if it satisfies

- $N(0) = 0$.
- **Independent increments:** For $0 \leq t_1 < t_2 < \dots < t_n$, the random variables

$$N(t_2) - N(t_1), N(t_3) - N(t_2), \dots$$



are independent.

- **Stationary increments:** The distribution of $N(t + s) - N(s)$ depends only on t .
- **Poisson increments:** For all $t \geq 0$,

$$N(t) \sim \text{Poisson}(\lambda t)$$



3 Markov Chain Theory

3.1 Basic Graph Theory

Definition. A **graph** is a pair

$$G = (V, E),$$

where

- V is a finite set of **vertices** (or nodes), and
- $E \subseteq \{\{i, j\} : i, j \in V, i \neq j\}$ is a set of **edges**.

Definition. Two vertices $i, j \in V$ are said to be **adjacent**, written

$$i \sim j,$$

if $\{i, j\} \in E$.

Definition. The **neighborhood** of a vertex i is

$$N(i) = \{j \in V : i \sim j\}$$

Definition. The degree of a vertex i , denoted $\deg(i)$, is the number of neighbors of i .

Definition. A **path** from vertex i to vertex j is a sequence

$$(i = v_0, v_1, \dots, v_n = j)$$

such that

$$v_{k-1} \sim v_k \quad \text{for all } k = 1, \dots, n$$

Definition. A graph is **connected** if for all $i, j \in V$, there exists a path from i to j .

Definition. A **cycle** is a path

$$(v_0, v_1, \dots, v_{n-1}, v_n = v_0)$$

with $n \geq 2$ and all vertices $v_0, \dots, v_{n-1}$ distinct.

Definition. A graph is **bipartite** if its vertex set can be written as

$$V = A \cup B, \quad A \cap B = \emptyset$$



such that every edge connects a vertex in A to a vertex in B .

3.2 Markov Chain

Definition. Let X_n represent the state of the system at time n . A sequence of random variables $\{X_n\}$ is a Markov chain if for all $n \geq 1$, the conditional probability of transitioning from state x_n to state x_{n+1} depends only on the current state x_n . That is,

$$\mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n, X_{n-1} = x_{n-1}, \dots, X_1 = x_1) = \mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n)$$

This is the Markov property, which can be written as

$$\mathbb{P}(X_{n+1} = x_{n+1} \mid \{X_n = x_n\}) = \mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n)$$

Definition. The behavior of a Markov chain is often represented using a **transition matrix** P . If the system has N possible states, the transition matrix is an $N \times N$ matrix where the entry p_{ij} represents the probability of transitioning from state i to state j . Mathematically,

$$P = \begin{pmatrix} p_{11} & p_{12} & \dots & p_{1N} \\ p_{21} & p_{22} & \dots & p_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ p_{N1} & p_{N2} & \dots & p_{NN} \end{pmatrix}$$

where each row sums to 1, i.e.,

$$\sum_{j=1}^N p_{ij} = 1 \quad \text{for all } i$$

Theorem. The (i, j) -entry of P^n is

$$p_{ij}^{(n)} = \mathbb{P}(X_n = j \mid X_0 = i)$$

Definition. A probability distribution

$$\boldsymbol{\pi} = (\pi_1, \pi_2, \dots, \pi_N)$$



is called a **stationary distribution** of a Markov chain with transition matrix P if

$$\pi P = \pi$$

that is

$$\pi_j = \sum_{i \in S} \pi_i p_{ij}, \quad \text{for all } j \in S$$

and

$$\sum_{j \in S} \pi_j = 1, \quad \pi_j \geq 0$$

Definition. State j is said to be **accessible** from state i , written

$$i \rightarrow j$$

if there exists some $n \geq 0$ such that

$$(P^n)_{ij} > 0$$

Definition. States i and j **communicate**, written

$$i \leftrightarrow j$$

if $i \rightarrow j$ and $j \rightarrow i$.

Definition. A Markov chain is called **irreducible** if for all $i, j \in S$,

$$i \leftrightarrow j$$

Equivalently, every state is accessible from every other state. Irreducibility means the chain is fully connected. There are no isolated regions of the state space: the chain can eventually move between any two states.

Definition. A probability distribution π is said to satisfy the **detailed balance condition** with respect to the transition matrix P if

$$\pi(i)P_{ij} = \pi(j)P_{ji}, \quad \forall i, j \in S$$

A Markov chain with stationary distribution π is called **reversible** if it satisfies the detailed balance equations.



Theorem. If a probability distribution π satisfies the detailed balance condition with respect to a transition matrix P , then π is a stationary distribution.

Proof. Fix $j \in S$. Using the detailed balance condition, we compute

$$\sum_{i \in S} \pi(i)P_{ij} = \sum_{i \in S} \pi(j)P_{ji} = \pi(j) \sum_{i \in S} P_{ji}$$

Since P is a transition matrix, each row sums to 1, so

$$\sum_{i \in S} P_{ji} = 1$$

Hence,

$$\sum_{i \in S} \pi(i)P_{ij} = \pi(j)$$

which shows that $\pi = \pi P$, and therefore π is stationary.

Theorem. Let $G = (V, E)$ be a finite, connected, undirected graph. Consider the simple random walk on G , where from a vertex i the walk moves to each neighboring vertex with equal probability. Then the stationary distribution π of the random walk is given by

$$\pi(i) = \frac{\deg(i)}{\sum_{v \in V} \deg(v)}, \quad i \in V$$

Proof. Let $|V| = n$. For the simple random walk on G , the transition probabilities are

$$P_{ij} = \begin{cases} \frac{1}{\deg(i)}, & \text{if } (i, j) \in E \\ 0, & \text{otherwise} \end{cases}$$

We claim that

$$\pi(i) = \frac{\deg(i)}{\sum_{v \in V} \deg(v)}$$

is a stationary distribution. To verify this, we use the **detailed balance condition**. For an undirected graph, if a distribution π satisfies

$$\pi(i)P_{ij} = \pi(j)P_{ji} \quad \text{for all } i, j \in V$$



then π is stationary. If $(i, j) \in E$, then

$$\pi(i)P_{ij} = \frac{\deg(i)}{\sum_{v \in V} \deg(v)} \cdot \frac{1}{\deg(i)} = \frac{1}{\sum_{v \in V} \deg(v)}$$

Similarly,

$$\pi(j)P_{ji} = \frac{\deg(j)}{\sum_{v \in V} \deg(v)} \cdot \frac{1}{\deg(j)} = \frac{1}{\sum_{v \in V} \deg(v)}$$

Hence,

$$\pi(i)P_{ij} = \pi(j)P_{ji}.$$

If $(i, j) \notin E$, then $P_{ij} = P_{ji} = 0$, and the detailed balance condition holds trivially. Therefore, π satisfies detailed balance with respect to P . Since π satisfies the detailed balance condition with respect to P , it follows that π is a stationary distribution. Finally, note that

$$\sum_{i \in V} \pi(i) = \frac{\sum_{i \in V} \deg(i)}{\sum_{v \in V} \deg(v)} = 1$$

so π is indeed a valid probability distribution.

Example. (Simple Random Walk) Let $\{X_n\}_{n \geq 0}$ be a stochastic process with state space

$$S = \mathbb{Z}$$

At each step, the process moves

- one unit to the right with probability p , or
- one unit to the left with probability $q = 1 - p$.

Given the current position $X_n = i$, the next position X_{n+1} depends only on i and not on how the process arrived at i . Hence,

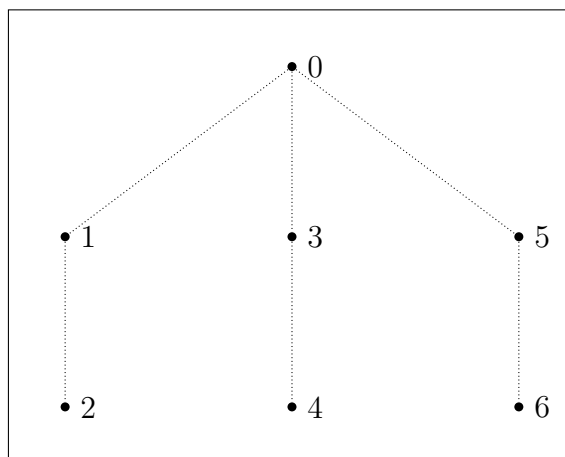
$$\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots, X_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i)$$

so the random walk satisfies the Markov property. For the simple random walk,

$$p_{ij} = \begin{cases} p, & j = i + 1 \\ q, & j = i - 1 \\ 0, & \text{otherwise} \end{cases}$$



Example.



Let $\{X_n\}_{n \in \mathbb{N}_0}$ be the random walk on the graph in Figure 1.

(a) Find the transition matrix of this Markov chain.

At each step, the random walk moves uniformly to one of the neighboring vertices. The degrees of the vertices are

$$\deg(0) = 3, \quad \deg(1) = 2, \quad \deg(5) = 1, \quad \deg(3) = 2, \quad \deg(2) = 1, \quad \deg(4) = 2, \quad \deg(6) = 1$$

The transition matrix $P = (p_{ij})$ is

$$P = \begin{pmatrix} 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

(b) Explain why this chain has a unique stationary distribution without finding the distribution.

- The graph is connected. Therefore, the Markov chain is irreducible.
- The state space is finite.

Therefore, the chain has a **unique stationary distribution**.



(c) Find the long-run proportion of time spent at each of the seven vertices.

For a simple random walk on an undirected graph, the stationary distribution satisfies

$$\pi(i) = \frac{\deg(i)}{\sum_{v \in V} \deg(v)}$$

The total degree is

$$\sum_{v \in V} \deg(v) = 3 + 2 + 1 + 2 + 1 + 2 + 1 = 12$$

Hence,

$$\pi(i) = \frac{\deg(i)}{12}$$

Therefore, the long-run proportion of time spent at each vertex is

Vertex	$\pi(i)$
0	$\frac{3}{12} = \frac{1}{4}$
1	$\frac{2}{12} = \frac{1}{6}$
2	$\frac{1}{12}$
3	$\frac{2}{12} = \frac{1}{6}$
4	$\frac{1}{12}$
5	$\frac{2}{12} = \frac{1}{6}$
6	$\frac{1}{12}$

Definition. Let $\{X_n\}_{n \geq 0}$ be a discrete-time Markov chain with state space S and transition matrix $P = (p_{ij})$. The **period** of a state $i \in S$ is defined as

$$d(i) = \gcd \left\{ n \geq 1 : p_{ii}^{(n)} > 0 \right\},$$

where $p_{ii}^{(n)} = \mathbb{P}(X_n = i \mid X_0 = i)$. A state i is called **aperiodic** if

$$d(i) = 1$$

A Markov chain is called **aperiodic** if all its states are aperiodic.

Definition. Let $\{X_n\}_{n \geq 0}$ be a discrete-time Markov chain with finite or countable state space S and



transition matrix $P = (p_{ij})$. If the limit

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n = j \mid X_0 = i)$$

exists and is independent of i , we denote it by

$$\pi(j) = \lim_{n \rightarrow \infty} p_{ij}^{(n)}$$

and call $\pi = (\pi(j))_{j \in S}$ the **limiting distribution**.

Theorem. Let $\{X_n\}_{n \geq 0}$ be a Markov chain with transition matrix P on a (finite or countable) state space S . If π is a **limiting distribution**, i.e.

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n = j \mid X_0 = i) = \pi(j) \quad \text{for all } i, j \in S$$

then π is a **stationary distribution**, that is,

$$\pi = \pi P$$

Proof. Fix any initial state $i \in S$. By the Markov property,

$$\mathbb{P}(X_{n+1} = j \mid X_0 = i) = \sum_{k \in S} \mathbb{P}(X_n = k \mid X_0 = i) P_{kj}$$

where $P_{kj} = \mathbb{P}(X_1 = j \mid X_0 = k)$. Taking the limit as $n \rightarrow \infty$, and using the assumption that

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n = k \mid X_0 = i) = \pi(k) \quad \text{for all } k \in S$$

we obtain

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_{n+1} = j \mid X_0 = i) = \sum_{k \in S} \pi(k) P_{kj}$$

On the other hand, since π is a limiting distribution, the left-hand side also equals $\pi(j)$. Hence, for every $j \in S$,

$$\pi(j) = \sum_{k \in S} \pi(k) P_{kj}$$

which is $\pi = \pi P$ in vector notation.



Part II: MATH3906 and MATH7224

4 Stochastic Calculus for MATH3906 and MATH7224

Definition. (Brownian Motion) A process $\{W_t\}_{t \geq 0}$ is called a **Brownian motion** if

- $W_0 = 0$ almost surely,
- $W_t - W_s \sim \mathcal{N}(0, t - s)$ for $t > s$,
- increments are independent, and
- paths are continuous almost surely.

Definition. The Itô integral of a stochastic process X_t with respect to Brownian motion W_t over the interval $[0, T]$ is defined as the limit in the mean-square sense:

$$\int_0^T X_t dW_t = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} X_{t_i} (W_{t_{i+1}} - W_{t_i})$$

Definition. A filtration on a probability space $(\Omega, \mathcal{F}, P)$ is a family of σ -algebras $\{\mathcal{F}_t\}_{t \geq 0}$ such that they are increasing:

$$\mathcal{F}_s \subseteq \mathcal{F}_t \quad \text{for all } s \leq t$$

A stochastic process X_t is said to be adapted to the filtration $\mathcal{F}_t$ if, for every t , the value of X_t is "known" based on the information in $\mathcal{F}_t$. Formally, X_t must be $\mathcal{F}_t$ -measurable. If W_t is a Brownian motion, its natural filtration is the one generated by its own history:

$$\mathcal{F}_t^W = \sigma(W_s : 0 \leq s \leq t)$$

W_t is automatically adapted to its natural filtration because you know where the particle is at time t if you have watched its entire path up to that moment.

Theorem. Let W_t be a standard Brownian motion and X_t be a square-integrable stochastic process adapted to the filtration $\mathcal{F}_t$. The Itô Isometry states:

$$\mathbb{E} \left[\left(\int_0^T X_t dW_t \right)^2 \right] = \mathbb{E} \left[\int_0^T X_t^2 dt \right] \quad (1)$$



Proof. Consider a partition of $[0, T]$ given by $0 = t_0 < t_1 < \cdots < t_n = T$. Let X_t be a simple process that is constant on each sub-interval:

$$X_t = X_{t_i} \quad \text{for } t \in [t_i, t_{i+1})$$

Consider

$$I(X) = \sum_{i=0}^{n-1} X_{t_i} \Delta W_i, \quad \text{where } \Delta W_i = W_{t_{i+1}} - W_{t_i}$$

Note that

$$\left(\sum_{i=0}^{n-1} X_{t_i} \Delta W_i \right)^2 = \sum_{i=0}^{n-1} X_{t_i}^2 (\Delta W_i)^2 + 2 \sum_{i < j} X_{t_i} X_{t_j} \Delta W_i \Delta W_j \quad (2)$$

By the linearity of expectation,

$$\mathbb{E}[I(X)^2] = \sum_{i=0}^{n-1} \mathbb{E}[X_{t_i}^2 (\Delta W_i)^2] + 2 \sum_{i < j} \mathbb{E}[X_{t_i} X_{t_j} \Delta W_i \Delta W_j] \quad (3)$$

Using the tower property, note that

$$\mathbb{E}[X_{t_i} X_{t_j} \Delta W_i \Delta W_j] = \mathbb{E}[\mathbb{E}[X_{t_i} X_{t_j} \Delta W_i \Delta W_j \mid \mathcal{F}_{t_j}]]$$

Since X_{t_i} , X_{t_j} , and ΔW_i are all $\mathcal{F}_{t_j}$ -measurable, they come out of the inner expectation:

$$\mathbb{E}[X_{t_i} X_{t_j} \Delta W_i \cdot \mathbb{E}[\Delta W_j \mid \mathcal{F}_{t_j}]]$$

As Brownian increments have zero mean ($\mathbb{E}[\Delta W_j] = 0$), the entire cross-term vanishes. Similarly,

$$\mathbb{E}[X_{t_i}^2 (\Delta W_i)^2] = \mathbb{E}[\mathbb{E}[X_{t_i}^2 (\Delta W_i)^2 \mid \mathcal{F}_{t_i}]]$$

As X_{t_i} is $\mathcal{F}_{t_i}$ -measurable:

$$\mathbb{E}[X_{t_i}^2 \cdot \mathbb{E}[(\Delta W_i)^2 \mid \mathcal{F}_{t_i}]]$$

By the properties of Brownian motion, $\mathbb{E}[(\Delta W_i)^2] = \Delta t_i$:

$$\mathbb{E}[X_{t_i}^2 \Delta t_i]$$



Summing these terms, we get

$$\mathbb{E}[I(X)^2] = \sum_{i=0}^{n-1} \mathbb{E}[X_{t_i}^2] \Delta t_i = \mathbb{E} \left[\sum_{i=0}^{n-1} X_{t_i}^2 \Delta t_i \right] \quad (4)$$

As the mesh of the partition goes to zero ($n \rightarrow \infty$), the sum converges to the Riemann integral:

$$\mathbb{E} \left[\left(\int_0^T X_t dW_t \right)^2 \right] = \mathbb{E} \left[\int_0^T X_t^2 dt \right] \quad (5)$$

Definition. A stochastic differential equation (SDE) is a differential equation in which one or more terms are driven by random noise. They are typically written in the form

$$dX(t) = \mu(t, X(t)) dt + \sigma(t, X(t)) dW(t)$$

where

- $X(t)$ is the state of the system at time t ,
- $\mu(t, X(t))$ is the drift term (deterministic part),
- $\sigma(t, X(t))$ is the diffusion term (stochastic part),
- $W(t)$ is a Wiener process or Brownian motion, representing the randomness.

Theorem. Let X_t be a stochastic process satisfying the stochastic differential equation (SDE)

$$dX_t = \mu(t)dt + \sigma(t)dW_t$$

where W_t is a standard Brownian motion, $\mu(t)$ is the drift term, and $\sigma(t)$ is the volatility term. If $f(t, X_t)$ is a sufficiently smooth function, then the differential of $f(t, X_t)$ is given by Itô's Lemma:

$$df(t, X_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} (dX_t)^2$$

Substituting the expression for dX_t , we get

$$df(t, X_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} (\mu(t)dt + \sigma(t)dW_t) + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} \sigma(t)^2 dt$$



Simplifying,

$$df(t, X_t) = \left(\frac{\partial f}{\partial t} + \mu(t) \frac{\partial f}{\partial x} + \frac{1}{2} \sigma(t)^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma(t) \frac{\partial f}{\partial x} dW_t$$

Proof. Consider a small increment Δt of the process X_t . The change in $f(t, X_t)$ can be approximated using a Taylor series expansion around t and X_t . Let $\Delta X_t = X_{t+\Delta t} - X_t$.

$$f(t + \Delta t, X_{t+\Delta t}) \approx f(t, X_t) + \frac{\partial f}{\partial t} \Delta t + \frac{\partial f}{\partial x} \Delta X_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} (\Delta X_t)^2 + \dots$$

As ΔX_t is the increment of a stochastic process,

$$\Delta X_t = \mu(t) \Delta t + \sigma(t) \Delta W_t$$

where ΔW_t is the increment of a Wiener process (Brownian motion). Substitute this expression for ΔX_t into the Taylor expansion:

$$f(t + \Delta t, X_{t+\Delta t}) \approx f(t, X_t) + \frac{\partial f}{\partial t} \Delta t + \frac{\partial f}{\partial x} (\mu(t) \Delta t + \sigma(t) \Delta W_t) + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} (\mu(t) \Delta t + \sigma(t) \Delta W_t)^2$$

$$f(t + \Delta t, X_{t+\Delta t}) \approx f(t, X_t) + \frac{\partial f}{\partial t} \Delta t + \mu(t) \frac{\partial f}{\partial x} \Delta t + \sigma(t) \frac{\partial f}{\partial x} \Delta W_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} (\sigma(t)^2 \Delta t)$$

Hence, by taking $\Delta t \rightarrow 0$,

$$df(t, X_t) = \left(\frac{\partial f}{\partial t} + \mu(t) \frac{\partial f}{\partial x} + \frac{1}{2} \sigma(t)^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma(t) \frac{\partial f}{\partial x} dW_t$$

Example. The equation

$$dS(t) = \mu S(t) dt + \sigma S(t) dW(t)$$

is a geometric Brownian motion (GBM). Using the Itô's Lemma for logarithmic transformation, define $X(t) = \ln S(t)$. Then, applying Itô's Lemma:

$$dX(t) = \frac{1}{S(t)} dS(t) - \frac{1}{2S(t)^2} (dS(t))^2$$

Substituting $dS(t) = \mu S(t) dt + \sigma S(t) dW(t)$, we get

$$dX(t) = \frac{1}{S(t)} (\mu S(t) dt + \sigma S(t) dW(t)) - \frac{1}{2S(t)^2} \sigma^2 S(t)^2 dt$$



Simplifying this:

$$dX(t) = \mu dt + \sigma dW(t) - \frac{1}{2}\sigma^2 dt$$

which gives the SDE for $X(t)$:

$$dX(t) = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dW(t)$$

This is a standard SDE for $X(t)$, which can be solved by integration:

$$X(t) = X(0) + \left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W(t)$$

Exponentiating both sides:

$$S(t) = S(0) \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W(t)\right)$$

5 Introduction to Financial Instruments for MATH3906

Definition. A stock (also called a share or equity) represents a claim on the assets and earnings of a company. There are two main types of stock:

- **Common stock:** Gives shareholders the right to vote on company matters and to receive dividends.
- **Preferred stock:** Typically does not give voting rights, but shareholders have a higher claim on assets and earnings than common stockholders.

The price of a stock is determined by supply and demand in the market. It fluctuates based on the company's performance, market conditions, and investor sentiment.

Definition. A bond is a debt security issued by an entity such as a corporation, government, or municipality to raise capital. When you purchase a bond, you are essentially lending money to the issuer in exchange for periodic interest payments and the return of the principal at maturity. Bonds are categorized as follows

- **Corporate bonds:** Issued by companies.
- **Government bonds:** Issued by governments.
- **Municipal bonds:** Issued by local governments.

The bond's price is influenced by factors like interest rates, credit ratings of the issuer, and the bond's maturity period. Bond prices and interest rates have an inverse relationship.

